Dataset & Research Objective

Forecasting DBS Group daily stock prices with a hybrid neuro-fuzzy model

Selected financial dataset

- Asset: DBS Group Holdings Ltd. (SGX ticker: D05.SI), Singapore-listed bank.
- Source: Yahoo Finance historical daily prices; proposed range: Jan 2020 - Dec 2025.
- Raw fields: Open, High, Low, Close, Adjusted Close (if available), Volume.
- Engineered features: daily returns, moving averages and rolling volatility.

Prediction task and processing

- Use a rolling 30-trading-day input window to predict the next trading day's closing price.
- Check missing observations, corporate actions and the consistency of price adjustments.
- Normalize features using training-set statistics only to prevent information leakage.
- Preserve chronological order when constructing training, validation and test sets.

Research questions

- Does fuzzy representation improve LSTM price-forecasting accuracy?
- Can model forecasts produce better simulated trading returns under common rules?

DATA ACCESS / HISTORICAL PRICES
https://sg.finance.yahoo.com/quote/D05.SI/history/

Hybrid AI Method: Fuzzy LSTM

Proposed design: fuzzy feature transformation followed by temporal forecasting

Architecture (proposed, subject to technical validation)

- Inputs: 30-day sequences of historical prices and selected technical indicators.
- Fuzzification: map selected market features (e.g., volatility/return) to low, medium and high memberships.
- Feature fusion: concatenate fuzzy memberships with standardized numeric inputs.
- LSTM layers learn temporal patterns; a dense output layer predicts next-day closing price.

Implementation plan

- Choose triangular or Gaussian membership functions, with ranges fitted on the training data.
- Train the model on past observations with a regression loss; tune parameters on validation data.
- Compare with ordinary LSTM using aligned data, horizons and evaluation settings.
- Run an ablation study: LSTM without fuzzy features versus LSTM with fuzzy features.

Research and reproducibility references

- Existing public notebooks cover LSTM baselines, fuzzy input representations and modified LSTM variants.
- Our final architecture will be aligned with literature and the verified notebook implementation.

PUBLIC RESEARCH CODE / FUZZY LSTM NOTEBOOKS
https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM

ASSOCIATED PUBLICATION / DOI LINK (VERIFY BIBLIOGRAPHIC DETAILS)
https://doi.org/10.1007/978-3-032-15761-4_4

Benchmark, Evaluation & Resources

Controlled comparison, backtesting and open-source starting points

Benchmark and evaluation

- Baseline: standard LSTM, trained on the same target, windows and financial dataset.
- Validation: chronological holdout plus walk-forward evaluation where feasible.
- Metrics: RMSE (primary), MAE; evaluate simulated trading returns versus buy-and-hold.
- Illustrative signal: invest only when predicted next-day return is positive; otherwise hold cash.

Planned deliverables and cautions

- Deliver a reproducible Jupyter Notebook plus literature review, results, analysis and slides.
- Assess transaction costs and look-ahead bias; improvements are hypotheses, not guaranteed.

LINKS TO INCLUDE IN FINAL SLIDES / IMPLEMENTATION NOTES

1 Yahoo Finance DBS historical dataset
https://sg.finance.yahoo.com/quote/D05.SI/history/

2 Fuzzy LSTM public notebooks / research code
https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM

3 Course sample Colab - downloading data / signals / portfolio
https://colab.research.google.com/drive/14VEWf5KzSxrTm8V4Px6cDu3eN3KUJXk5